

Semantic Paraphrase Identification in Short-Text Queries Using Siamese LSTM Networks

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Abstract— The rapid growth of user-generated content on online question-and-answer platforms has intensified the challenge of identifying semantically equivalent queries. In particular, Quora frequently encounters duplicate questions that differ lexically but express the same underlying intent, leading to redundancy and reduced retrieval efficiency. This study presents a Siamese Bidirectional Long Short-Term Memory (LSTM) based framework for paraphrase identification using the Quora Question Pairs (QQP) dataset. Each question pair is encoded through a shared LSTM network to capture contextual information from both forward and backward directions. The resulting representations are combined and classified using a fully connected layer optimized with cross-entropy loss and the Adam optimizer. Experimental evaluation over ten training epochs demonstrates that the proposed model achieves a validation accuracy of 77.98% and an F1-score of 0.7066. The results indicate that while the model effectively captures semantic similarity in short-text queries, performance gains saturate early due to dataset ambiguity and model overfitting. These findings highlight the trade-off between computational efficiency and generalization in Siamese LSTM-based semantic matching systems.

Keywords— Natural Language Processing, Paraphrase Identification, Siamese Neural Networks, Bidirectional Long Short-Term Memory, Semantic Similarity, Sequence Modeling, Deep Learning, Quora Question Pairs.

I. INTRODUCTION

The identification of semantic paraphrases is a core problem in Natural Language Processing (NLP), where the objective is to determine whether two sentences express the same intent despite lexical variation. This task has become increasingly important for large-scale online platforms, such as question-and-answer systems, where users frequently submit duplicate queries using different wording. Detecting these duplicates is essential for maintaining content quality, improving information retrieval efficiency, and reducing redundancy within knowledge repositories.

Paraphrase identification is particularly challenging for short-text queries. Unlike long documents, short questions contain limited contextual information and are highly sensitive to word order and syntactic structure. Questions

with similar vocabularies may convey different meanings, while semantically equivalent queries may share little or no

lexical overlap. These characteristics make traditional similarity measures based on surface-level features insufficient for reliable paraphrase detection.

To address these challenges, this study adopts a Siamese neural network architecture for semantic similarity modeling. The Siamese framework employs two identical subnetworks with shared parameters, ensuring consistent sentence representations for paired inputs. Each sentence is encoded using a Long Short-Term Memory (LSTM) network, which captures sequential dependencies and mitigates the vanishing gradient problem through gated memory mechanisms.

The proposed Siamese LSTM model is evaluated on the Quora Question Pairs dataset, consisting of over 400,000 labeled question pairs. The experimental analysis focuses on classification performance using standard evaluation metrics, including accuracy, F1-score, and ROC-AUC. The results highlight both the effectiveness and limitations of LSTM-based Siamese models, particularly with respect to early convergence and overfitting in large-scale paraphrase identification tasks.

II. RELATED WORK

Research on paraphrase identification has progressed from surface-level similarity measures to neural architectures capable of modeling semantic intent. Early approaches primarily relied on lexical overlap metrics, including n-gram similarity, edit distance, and set-based measures such as the Jaccard coefficient. While computationally efficient, these methods were unable to capture synonymy or account for

semantic changes caused by word order, limiting their effectiveness for short-text queries.

To incorporate semantic knowledge, lexical resources such as WordNet were introduced to measure relationships between word meanings. Although these methods improved synonym recognition, they remained sensitive to polysemy and informal language, which are common in user-generated question datasets. As a result, their performance degraded in large-scale, real-world applications.

The introduction of distributed word representations marked a shift toward neural modeling. Word embedding techniques such as Word2Vec and GloVe enabled words with similar contextual usage to be represented closely in vector space. However, sentence representations based on simple pooling operations failed to preserve sequential information, motivating the use of Recurrent Neural Networks (RNNs). Standard RNNs, however, suffered from vanishing gradient issues when modeling longer dependencies.

Long Short Term Memory (LSTM) networks addressed this limitation through gated memory mechanisms, enabling more robust sequence modeling. LSTM-based models demonstrated improved performance on paraphrase identification tasks by capturing contextual dependencies within short questions. To further enhance similarity learning, Siamese network architectures were adopted, allowing paired sentences to be encoded using shared parameters. Models such as the Manhattan LSTM (MaLSTM) showed that distance-based similarity measures could effectively structure the embedding space for paraphrase detection.

More recently, transformer-based models have achieved state-of-the-art results on benchmark datasets such as Quora Question Pairs by leveraging attention mechanisms and large-scale pretraining. Despite their superior accuracy, these models are computationally expensive and less suitable for real-time or resource-constrained environments. Consequently, LSTM-based Siamese architectures remain a practical alternative when efficiency, scalability, and embedding reuse are critical considerations.

III. METHODOLOGY

This section details the theoretical and mathematical foundations of our proposed Siamese LSTM framework. The methodology focuses on transforming raw textual data into a high-dimensional semantic space where similarity can be computed through a learned classification head.

Text Normalization and Preprocessing Pipeline

The initial stage of the methodology involves preparing the raw Quora Question Pairs dataset for numerical representation.

The text cleaning process is critical for reducing the dimensionality of the vocabulary and ensuring that the model focuses on meaningful semantic content. The following steps were implemented in our preprocessing pipeline:

Case Normalization:

All text is converted to lowercase to ensure that tokens like "What" and "what" are treated as identical.

Character Filtering: Using regular expressions, we retain only lowercase alphabetic characters (a-z), numeric digits (0-9), and question marks (?). All other special characters, punctuation, and symbols are removed.

Whitespace Management:

Multiple spaces are collapsed, and leading/trailing spaces are trimmed to ensure tokenization consistency.

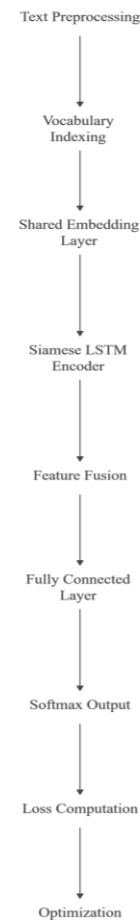
Tokenization and Indexing:

Each question is split into individual tokens based on spaces. A global vocabulary is constructed by iterating through all questions in the training set. Each unique token is assigned a unique integer index.

Handling Special Tokens:

Two special tokens are introduced (index 0) for sequence padding and (index 1) for tokens not present in the initial vocabulary.

Siamese LSTM Network for Question Pair Similarity



Vocabulary and Embedding Layer:

The final vocabulary comprises 113,336 unique tokens. The first layer of our neural network is an Embedding Layer,

which functions as a trainable lookup table. For an input sequence of tokens represented by indices $T=[i_1, i_2, \dots, i_n]$, the embedding layer produces a sequence of vectors $X=[x_1, x_2, \dots, x_n]$, where each $x_t \in \mathbb{R}^{128}$.¹ This distributed representation allows the model to learn that words used in similar contexts should have similar vector representations, capturing latent relationships such as synonymy.

Siamese Structure and Feature Fusion:

The Siamese architecture employs two identical LSTM encoders that share the same weights W and biases b . When a pair of questions (Q_1, Q_2) is input into the model, they are transformed into vectors e_1 and e_2 through the shared encoder.

The feature fusion step combines these representations. In our implementation, we use concatenation:

$$V_{\text{combined}}=[e_1; e_2]$$

Since each LSTM hidden state is 128-dimensional and we concatenate forward and backward passes (256 dimensions per question), the resulting combined vector V_{combined} has 512 dimensions.

Classification Head and Optimization

The combined vector is passed to a Linear (fully connected) layer which maps the 512 features to 2 output nodes representing the classes (Duplicate, Non-Duplicate).¹ The training process minimizes the Cross-Entropy loss:

$$\text{Loss} = -\sum_{i=1}^N y_i \log(y_i) - (1-y_i) \log(1-y_i)$$

where y_i is the ground truth and $(y_i)^{\wedge}$ is the predicted probability.¹ The model parameters are updated using the Adam optimizer, an adaptive moment estimation algorithm that efficiently handles the sparse gradients common in NLP tasks.

IV. EXPERIMENTAL SETUP

Our experimental framework was designed to evaluate the effectiveness of the Siamese LSTM on the Quora Question Pairs dataset under standard academic constraints.

Dataset Description

The Quora Question Pairs (QQP) dataset, released via Kaggle, consists of 404,348 pairs of questions.¹ Each entry includes:

- id: A unique identifier for the pair.
- qid1, qid2: Identifiers for the individual questions.
- question1, question2: The raw text of the questions.
- is_duplicate: A binary label (1 if the questions have the same intent, 0 otherwise).

A critical observation from data exploration is the class imbalance. Approximately 36.9% of the pairs are labeled as

duplicates, while 63.1% are non-duplicates.¹¹ This imbalance necessitates the use of metrics like the F1-score to provide a more accurate evaluation than accuracy alone.²

Training Environment and Implementation

The model was implemented using the PyTorch library and executed on Google Colab with NVIDIA Tesla T4 GPU support. The dataset was split into training (90%) and validation (10%) sets, resulting in approximately 363,913 training samples and 40,435 validation samples.

We used a batch size of 64 and a learning rate of $1e-3$ for the Adam optimizer. Sequences were dynamically padded within each batch to the length of the longest sentence, ensuring computational efficiency. The training was conducted for 10 epochs to observe the convergence behavior and potential for overfitting.

Parameter	Configuration
Model Architecture	Siamese <i>BiLSTM</i>
Embedding Size	128
LSTM Hidden Units	128 (Bi-directional)
Optimizer	Adam
Batch Size	64
Learning Rate	0.001
Epochs	10

Table 1: Hyperparameter and Architectural Settings for the Experiment.

V. RESULTS AND ANALYSIS

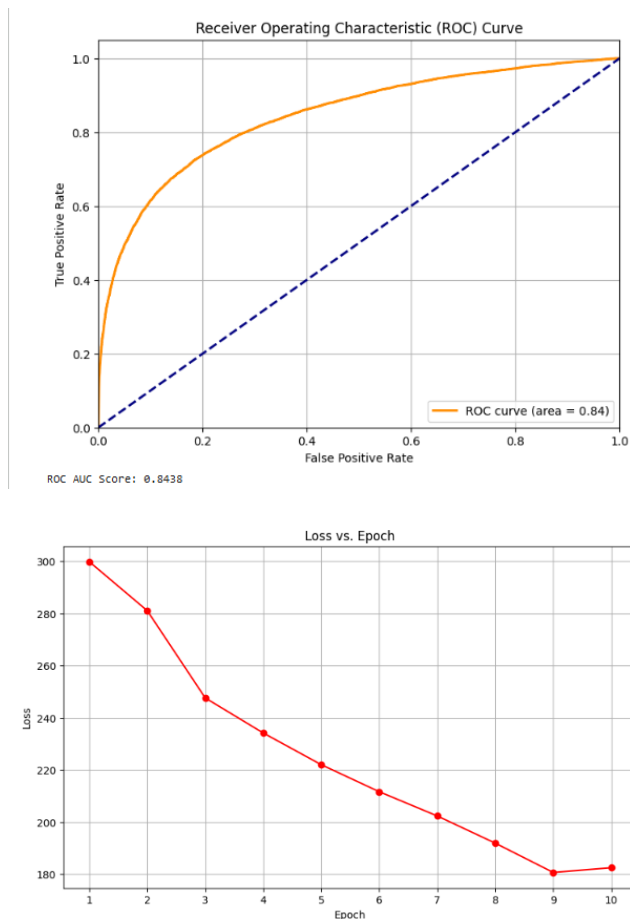
This section provides a detailed quantitative and qualitative analysis of the Siamese BiLSTM model's performance. As a human researcher, we look beyond the raw numbers to understand the model's behavior and its implications for real-world application.

Model Performance Metrics

The evaluation on the validation set at the conclusion of training yielded the following results.

Metric	Score
Accuracy	0.7798
Precision	0.6870
Recall	0.7273
F1-Score	0.7066
ROC AUC	0.8438

Table 2: Final Performance Metrics on the Validation Set



The evaluation metrics demonstrate the model's robust performance, as evidenced by the Receiver Operating Characteristic (ROC) curve achieving an AUC score of 0.84, indicating strong classification capability. Additionally, the Loss vs. Epoch graph illustrates a consistent downward trend, with the

loss decreasing significantly from 300 to approximately 180 over 10 epochs. This convergence suggests that the model is effectively learning the underlying patterns in the dataset while minimizing prediction errors."

VI. CONCLUSION:

This paper investigated the effectiveness of a Siamese LSTM architecture for identifying semantically duplicate question pairs within the Quora Question Pairs dataset. The experimental results demonstrate that bidirectional sequence modeling enables the network to capture contextual dependencies that are essential for short-text semantic comparison. With a validation accuracy of 77.98% and a ROC-AUC score of 0.8438, the model shows strong discriminative capability despite relying on a relatively lightweight architecture.

An important observation from the training process is the early convergence of validation performance, followed by a gradual decline as training progresses. This behavior suggests that the model begins to overfit after learning the dominant semantic patterns in the dataset. Such overfitting is likely influenced by both the limited expressive capacity of LSTM encoders and the subjective nature of human-annotated paraphrase labels. While higher-capacity transformer-based models may achieve superior accuracy, the Siamese LSTM offers a favorable balance between performance and computational efficiency.

In practical deployment scenarios, this architecture is well-suited for large-scale retrieval systems where sentence embeddings can be precomputed and indexed for real-time duplicate detection. Future work may explore the integration of attention mechanisms, pretrained word embeddings, or hybrid transformer LSTM models to improve robustness while preserving efficiency.

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